1 The four matrices

($e :=$ true $-$ estimate, $\mathbf{e} = [e_{\delta\bar{w}_1},\ e_{\delta\bar{w}_2},\ e_{\delta\bar{w}_3},\ e_{\bar{a}_w},\ e_p,\ e_\delta]^\top$; $\hat{g} = \bar{w}_d - \delta\hat{\bar{w}}_3 - \bar{w}_s + \hat{\delta}$, $\bar{a}' = \hat{g}^{-\hat{p}}$; all matrices evaluated at the estimate, step k)

$$\mathbf{e}[k+1] = F_e[k]\,\mathbf{e}[k] + \mathbf{g}[k]\,\varepsilon_w[k]$$

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} & 0 & 0 \\ 0 & 0 & \bar{a}'(1-\lambda_c) + \dfrac{\hat{p}\,\bar{a}'}{\hat{g}}M & 1 + \bar{a}' F_{dw} & -\bar{a}' \ln \hat{g}\, M & -\dfrac{\hat{p}\,\bar{a}'}{\hat{g}}M \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\big((\delta, c):\ F_{e,45} = -\tfrac{M}{\hat{g}^2},\ F_{e,46} = -\tfrac{2\bar{a}'}{\hat{g}}M\big)$$

$$\mathbf{y}[k] - \hat{\mathbf{y}}[k] = H[k]\,\mathbf{e}[k] + \mathbf{n}[k]$$

$$H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\dfrac{\hat{p}\,\bar{a}'}{\hat{g}}\nabla_d \bar{w}_d & 1 & \bar{a}' \ln \hat{g}\, \nabla_d \bar{w}_d & \dfrac{\hat{p}\,\bar{a}'}{\hat{g}}\nabla_d \bar{w}_d \end{bmatrix}$$

$$\big((\delta, c):\ H_{25} = \tfrac{\nabla_d \bar{w}_d}{\hat{g}^2},\ H_{26} = \tfrac{2\bar{a}'}{\hat{g}}\nabla_d \bar{w}_d\big)$$

$$Q[k] = Q_{33}[k]\,\mathbf{g}\mathbf{g}^\top, \qquad \mathbf{g} = [\,0,\ 0,\ -1,\ \bar{a}',\ 0,\ 0\,]^\top, \qquad Q_{33} = \kappa_T\,\hat{\bar{a}}_w$$

$$R[k] = \mathrm{diag}\big(R_1,\ R_2[k]\big), \qquad R_1 = \sigma_n^2 / R^2$$

2 The innovation as a linear equation in the start error

$$\Phi(k, k_0) := F_e[k-1]\,F_e[k-2]\cdots F_e[k_0], \qquad \Phi(k_0, k_0) = I$$

$$\mathbf{e}[k] = \Phi(k, k_0)\,\mathbf{e}[k_0] + \sum_{j=k_0}^{k-1} \Phi(k, j+1)\,\mathbf{g}[j]\,\varepsilon_w[j]$$

$$\mathbf{y}[k] - \hat{\mathbf{y}}[k] = \underbrace{H[k]\,\Phi(k, k_0)}_{\text{coefficients of } \mathbf{e}[k_0]}\ \mathbf{e}[k_0]\ +\ \mathbf{n}[k]\ +\ \underbrace{H[k] \sum_j \Phi(k, j+1)\,\mathbf{g}[j]\,\varepsilon_w[j]}_{\text{the } Q \text{ term (section 7)}}$$

(for a constant $\theta \in \{p, \delta\}$ the coefficient of e_θ in row y_i at step k is $h^{(i)}_{k,\theta} := [H[k]\Phi(k, k_0)]_{i\theta}$; along a descent $[\Phi(k, k_0)]_{4\theta} \approx \sum_j F_{e,4\theta}[j]$, the constant's error accumulated into the level)

3 Stacked system and least squares

(window $k = k_0, \ldots, k_0 + N$; the Q term dropped here, see section 7)

$$\mathbf{Y} := \begin{bmatrix} \mathbf{y}[k_0] - \hat{\mathbf{y}}[k_0] \\ \vdots \\ \mathbf{y}[k_0+N] - \hat{\mathbf{y}}[k_0+N] \end{bmatrix}, \qquad \mathbf{N} := \begin{bmatrix} \mathbf{n}[k_0] \\ \vdots \\ \mathbf{n}[k_0+N] \end{bmatrix}$$

$$O := \begin{bmatrix} H[k_0]\,\Phi(k_0,k_0) \\ \vdots \\ H[k_0+N]\,\Phi(k_0+N,k_0) \end{bmatrix}$$

$$\mathbf{Y} = O\,\mathbf{e}[k_0] + \mathbf{N}$$

$$\mathrm{Cov}(\mathbf{N}) = \mathcal{R} := \mathrm{diag}\big(R[k_0],\dots,R[k_0+N]\big)$$

$$\begin{aligned} \hat{\mathbf{e}}[k_0] &= \big(O^\top \mathcal{R}^{-1} O\big)^{-1} O^\top \mathcal{R}^{-1}\,\mathbf{Y} \\ &= \mathbf{e}[k_0] + \big(O^\top \mathcal{R}^{-1} O\big)^{-1} O^\top \mathcal{R}^{-1}\,\mathbf{N} \end{aligned}$$

$$\mathrm{Cov}\big(\hat{\mathbf{e}}[k_0] - \mathbf{e}[k_0]\big) = \big(O^\top \mathcal{R}^{-1} O\big)^{-1} =: G^{-1}$$

(the error covariance does not depend on $\mathbf{Y}$: the check needs only O and $\mathcal{R}$, i.e. the logged F_e, H, R)

4 Whitening, singular values, CRLB

$$O_w := \mathcal{R}^{-1/2}\,O \qquad \text{(each row divided by its own } \sqrt{R}\text{; every row's noise now has unit variance)}$$

$$G = O_w^\top O_w = \sum_k \Phi(k,k_0)^\top H[k]^\top R[k]^{-1} H[k]\,\Phi(k,k_0)$$

$$O_w\,\mathbf{v}_i = \sigma_i\,\mathbf{u}_i, \qquad G\,\mathbf{v}_i = \sigma_i^2\,\mathbf{v}_i$$

$$G^{-1} = \sum_i \frac{\mathbf{v}_i \mathbf{v}_i^\top}{\sigma_i^2} \qquad \text{(eigen-decomposition, not pinv: a } \sigma_i = 0 \text{ direction is } +\infty\text{)}$$

$$\mathrm{std}\big(\mathbf{v}_i^\top \hat{\mathbf{e}}[k_0]\big) = \frac{1}{\sigma_i} \qquad \text{(error bar along the } i\text{-th direction)}$$

$$\mathrm{CRLB}_\theta := \sqrt{[G^{-1}]_{\theta\theta}} = \sqrt{\sum_i \frac{(v_{i,\theta})^2}{\sigma_i^2}} \qquad \text{(slot } \theta\text{)}$$

(one unknown, one sensor, constant coefficient h over N steps: $\sigma = \sqrt{N}\,h/\sqrt{R}$, $\mathrm{CRLB} = \sqrt{R}/(h\sqrt{N})$)

5 Two constants: the alias direction

(the (e_p, e_δ) block of O_w at one height $\hat{g}$: every row is the same direction up to the scalar M or $\nabla_d \bar{w}_d$)

row direction at $\hat{g}$ $\propto$ $\mathbf{v}(\hat{g}) = \big[\ln\hat{g},\ \hat{p}/\hat{g}\,\big]$ $\qquad \big((\delta, c)\text{: } [\,1/\hat{g}^2,\ 2(1-\hat{c})/\hat{g}^3\,]\big)$

one height only $\Rightarrow \operatorname{rank}\big(O_w[:,(p,\delta)]\big) = 1, \quad \sigma_2 = 0$

only $\ln\hat{g}\, e_p + \frac{\hat{p}}{\hat{g}} e_\delta$ is determined

two heights $\Rightarrow \det\begin{bmatrix}\mathbf{v}(\hat{g}_1)\\ \mathbf{v}(\hat{g}_2)\end{bmatrix} = \dfrac{\hat{p}\,(\hat{g}_1\ln\hat{g}_1 - \hat{g}_2\ln\hat{g}_2)}{\hat{g}_1\hat{g}_2} \neq 0$

$\mathbf{v}_2$ = the eigenvector of the smallest σ (the weakly determined combination)

CRLB_θ is large for the slot with the larger $|v_{2,\theta}|$

6 Prior and the ratio

$\sigma_{\theta,0} := \sqrt{P_{\theta\theta}[0]}$ $\qquad$ (width believed before any data)

$$\frac{1}{\sigma^2_{\theta,\mathrm{post}}} = \frac{1}{\sigma^2_{\theta,0}} + \frac{1}{\mathrm{CRLB}^2_\theta}$$

$$\rho_\theta := \frac{\mathrm{CRLB}_\theta}{\sigma_{\theta,0}} \qquad \begin{cases} \rho \ll 1: & \sigma_{\mathrm{post}} \approx \mathrm{CRLB},\ \text{the data replace the belief} \\ \rho \gg 1: & \sigma_{\mathrm{post}} \approx \sigma_0,\ \text{the belief stays} \\ & \text{(observable in form, not identifiable in practice)} \\ \rho = \infty: & \text{no row of the window carries } e_\theta \end{cases}$$

7 What the omitted Q term does

$$\mathbf{Y} = O\,\mathbf{e}[k_0] + \mathbf{N} + \mathbf{W}$$

$$\mathbf{W}_k = H[k] \sum_{j=k_0}^{k-1} \Phi(k, j+1)\,\mathbf{g}[j]\,\varepsilon_w[j], \qquad \mathrm{Cov}(\mathbf{W}) =: \mathcal{Q}\ \text{(not diagonal)}$$

$G = O^\top \mathcal{R}^{-1} O$ $\qquad$ (the tool: $\mathcal{Q}$ dropped, structural question, optimistic)

$G_{\mathrm{full}} = O^\top (\mathcal{R} + \mathcal{Q})^{-1} O$ $\qquad$ (the same computed recursively is the filter's $P^- = F_e P F_e^\top + Q$)

$$\mathrm{honesty}_\theta := \frac{\mathrm{sd}_{\mathrm{seeds}}(\hat{\theta})}{\sqrt{P_{\theta\theta}}}, \qquad \mathrm{efficiency}_\theta := \frac{\sqrt{P_{\theta\theta}}}{\mathrm{CRLB}_\theta}$$

8 The five gates as formulas

gate 1 : $F_{e,4\theta}[k] \not\equiv 0$ or $H_{2\theta}[k] \not\equiv 0$

gate 2 : $h^{(i)}_{k,\theta} \not\propto h^{(i)}_{k,\theta'}$ as sequences in k (θ' an existing state)

gate 3 : $\det \begin{bmatrix} C_2 \\ C_2 A_2 \end{bmatrix} = 0$ solved for the operating condition (hold, far field, one height)

gate 4 : $\operatorname{rank}\big(O_w[:, \text{free}]\big) = |\text{free}|, \qquad \text{CRLB}_{\text{neg ctrl}} = +\infty$ (a locked slot: its column of O is zero)

gate 5 : $\rho_\theta < 1$ in at least one window; $\rho^{(y_1)}_\theta$, $\rho^{(y_2)}_\theta$ (rows of one sensor only); ρ_θ per segment

9 Constant-velocity descent

($\bar{w}_d$ from $\bar{w}_{\text{top}}$ to $\bar{w}_{\text{bot}}$ in T_d; $v = (\bar{w}_{\text{top}} - \bar{w}_{\text{bot}})/T_d$; $M = v\,T_s$; steps $N = T_d/T_s$; hold at both ends)

$$\text{hold} : M = 0,\ \nabla_d \bar{w}_d = 0 \ \Rightarrow\ F_{e,4\theta} = H_{2\theta} = 0 \ \Rightarrow\ \rho_\theta = \infty$$

$$\text{row directions} : \mathbf{v}(\hat{g}) \text{ depends on the heights visited only}$$

$$\Rightarrow \text{ the separation of } (p, \delta) \text{ does not depend on } T_d$$

$$y_2 \text{ rows} : h^{(2)}_{k,\theta} = [\Phi(k, k_0)]_{4\theta} \text{ (no } F_{dw}), \qquad \sum_k \frac{(h^{(2)}_{k,\theta})^2}{R_2} \propto N \propto T_d$$

$$y_1 \text{ rows} : h^{(1)}_{k,\theta} \approx \frac{-F_{dw}}{1 - \lambda_c}\,[\Phi(k, k_0)]_{4\theta}, \quad F_{dw} = \frac{M}{\hat{\bar{a}}_w} \propto \frac{1}{T_d}$$

$$\sum_k \frac{(h^{(1)}_{k,\theta})^2}{R_1} \propto \frac{1}{T_d^2}\,T_d = \frac{1}{T_d}$$

$$\sigma^2_\theta(T_d) \approx \frac{A_\theta}{T_d} + B_\theta\,T_d, \qquad \text{CRLB}_\theta(T_d) = \frac{1}{\sigma_\theta(T_d)} \ \text{ is largest at } T_d^* = \sqrt{A_\theta / B_\theta}$$

$$\text{forcing of a wrong slope} : \int \big(\bar{a}'_{\text{true}} - \bar{a}'\big)\,dh \quad \text{(independent of } T_d)$$

correction by $y_2 \propto T_d$